



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 24: Pyro Sensor Bi-Weekly Update 5

Alfonso Wong

Martin Ford

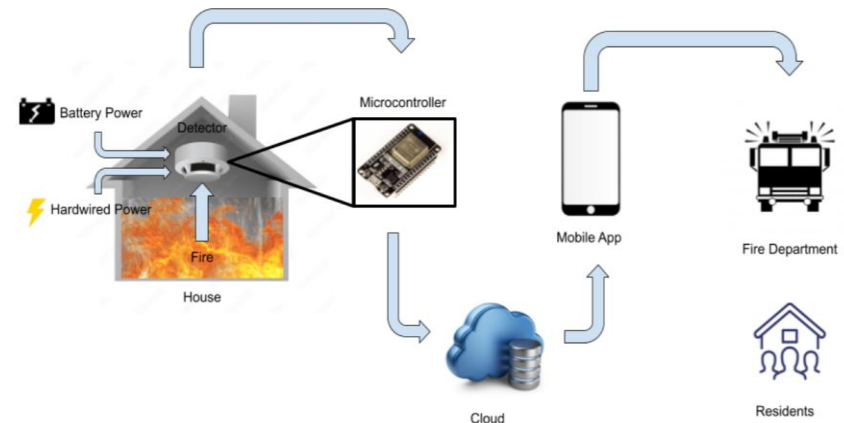
Oscar Hernandez

Sponsor: Rhett Guthrie

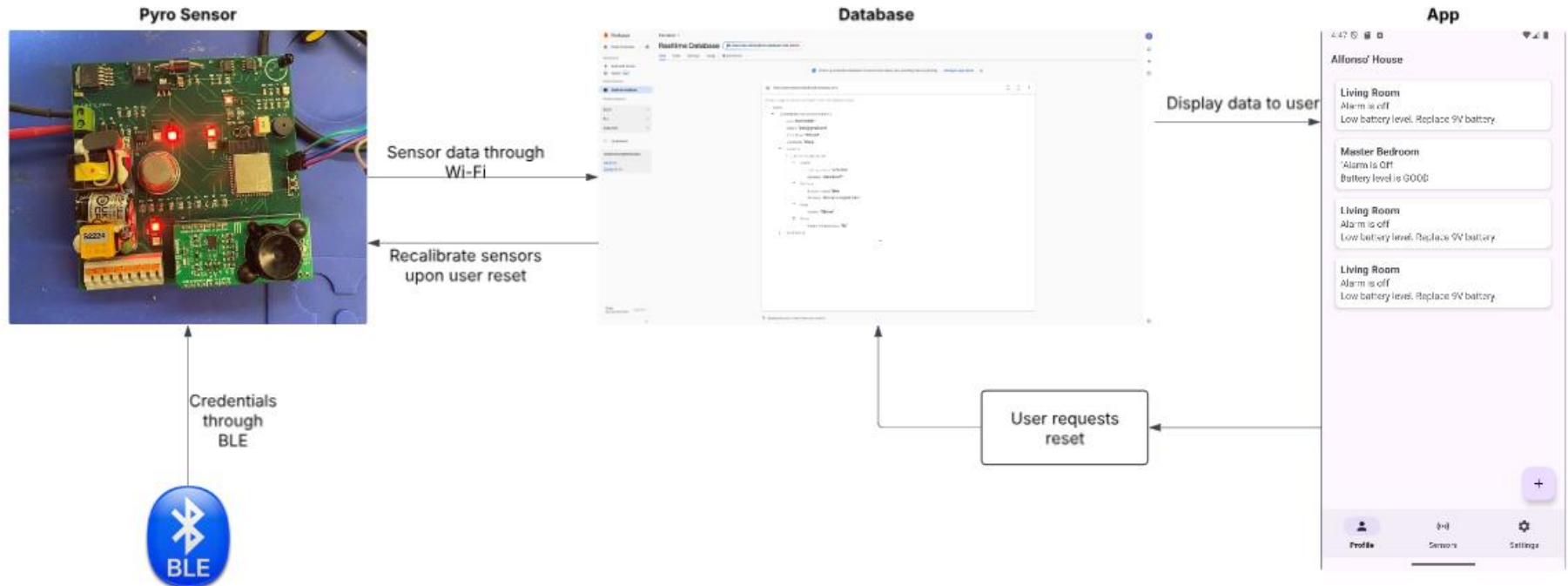
TA: Niloofar Borzooei

Project Summary

- Problem statement:
 - Modern smoke detectors do not directly notify the emergency responders
 - Fire department has limited data of residents in buildings
- Our Pyro Sensor will:
 - Notify any user and emergency responders instantly through a mobile app of any incident
 - Give fire department information about residents and pets



Integrated System Diagram





Project Timeline

Integration of mobile app and MCU (completed 2/28)	Subsystem Designs and Testing (completed 3/1)	Integration of Hardware and MCU (completed 3/2)	Final Integration (completed 3/2)	Full System Test (completed 3/22)	Validation (to complete by 4/3)	Demo and Report (to complete by 4/24)
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Martin Ford

Accomplishments since last update 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
Validated and tested full integrated system from signal reception to app notification	- Ongoing validation for different scenarios - Different levels/distances of IR, UV, smoke exposure



Oscar Hernandez

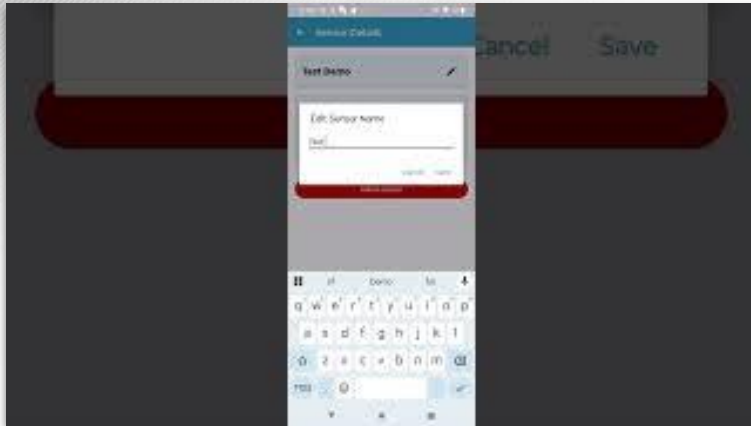
Accomplishments since last update 17 hrs of effort	Ongoing progress/problems and plans until the next presentation
- Validated correct system response to changes in the inputs. “Safe” notification in app for normal conditions. “Warning” for potential hazard or low battery level. “Alarm” notification for detected fire, and activate device buzzer. - Tested that reset command from user shuts off buzzer (if active) and recalibrates sensors.	- Developing scenarios to verify system stability during Wi-Fi loss and power outages. - On-going testing to verify that the device prevents going into Alarm mode from false alarm sources.



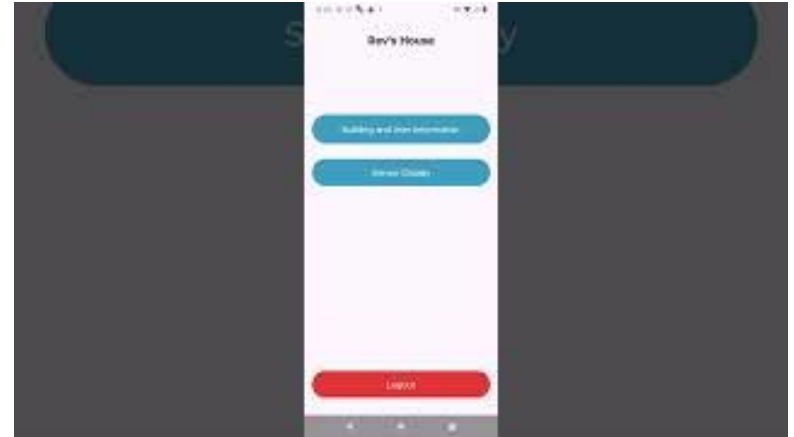
Alfonso Wong

Accomplishments since last update 16 hrs of effort	Ongoing progress/problems and plans until the next presentation
- Validated correct system response to changes in the inputs. “Safe” notification in app for normal conditions. “Warning” for potential hazard or low battery level. “Alarm” notification for detected fire - Validated alert text notification to central dispatcher - Validated firefighter sign in and sensor view - Validated sensor connection to mobile app .	- Need to validate picture uploads - Fire time-tracker is having some errors. Need to fix and validate

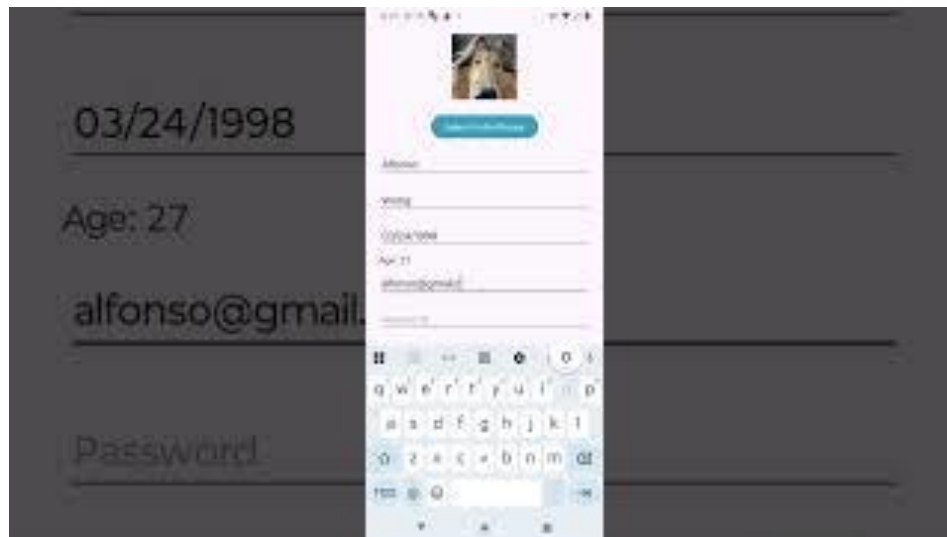
Mobile App Interface



User Login Demo

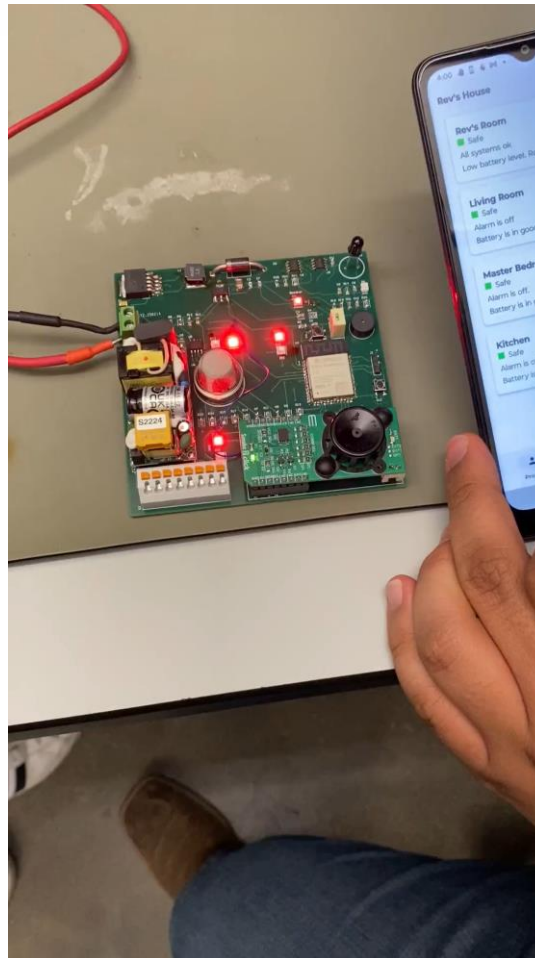


Firefighter Login Demo



User Account Creation Demo

Device Initialization and Real-Time Fire Response





Execution Plan

	12/03	02/03	02/17	03/03	03/24	04/07	Date
App's Major Functions Working							12/03
PCB tested and partially functional							12/03
Microcontroller Optimization							12/03
Connect MCU and app subsystems' databases							02/03
Fix notification System and picture uplaod							02/03
Switch communication to centralized dispatcher							02/03
PCB schematic fix							02/03
Testing succesful data package transfer for dispatcher							02/03
Begin integration							02/17
Download code to microcontroller							02/17
MCU and mobile app communicate in proper format							02/17
PCB reorder and resolder							03/03
PCB final testing							03/03
Connetion between MC and mobile app from app side							03/03
Stress test scenarios for functionality:							03/03
Data transfer and analysis from sensors to MCU							03/03
MCU notification of fire to mobile app							03/03
MCU transmits live data feed to mobile app							03/03
Full stability and consistency testing							03/24
Final validation							04/07

	Not Started		Hardware Subsystem
	In Progress		Microcontroller Subsystem
	Completed		Mobile App Subsystem
	Behind Schedule		Entire Team



Validation Plan

Paragraph #	Test Name	Success Criteria	Status	Responsible Engineers
3.2.1.1	Sensor Functionality	Sensors detect smoke/flame at expected ranges and output a signal.	Tested	Martin Ford
3.2.2.6	Speaker Functionality	Speaker emits sound from signal at rated buzzer volume output.	Tested	Martin Ford
3.2.2.5	Operating Time	System operates and receives full power continuously for 1 hour on each power source.	Untested	Martin Ford
3.2.1.4	Network	MCU is able to aggregate data from sensors to a Access Point.	Tested	Oscar Hernandez
3.2.1.14	Trouble Signal	MCU sends trouble signal to app for malfunctions.	Tested	Oscar Hernandez
3.2.1.8	User Profile Creation	User is able to create an account and be able to sign in.	Tested	Alfonso Wong
3.2.1.9	Push Notifications	Be able to push notifications to users and send text to central dispatcher.	Tested	Alfonso Wong
3.2.1.10	Design UI	Have a working UI and test with emulator.	Tested	Alfonso Wong
3.2.1.11	Integrate Cloud Service	Mobile app has ability to pull data from cloud service and display it.	Tested	Alfonso Wong



Validation Plan

Paragraph #	Test Name	Success Criteria	Status	Responsible Engineers
3.2.1.6	App compatible on Android devices	App can be used on most Android devices	Tested	Alfonso Wong
3.2.1.1	MCU Peripherals Interface	MCU reads signals from all sensors, and activates/deactivates alarm.	Tested	Martin Ford & Oscar Hernandez
3.2.1.12	Device and Application Communication	A new device can be added to user profile in application.	Tested	Alfonso Wong & Oscar Hernandez
3.2.2.13	Real Time Communication	MCU is feeding constant information to database and it is being displayed in real time in mobile application.	Tested	Alfonso Wong & Oscar Hernandez
3.2.1.9	Trigger Warning from MC	MCU must send a trigger alert of ongoing event and application must send notification to user and dispatch	Tested	Alfonso Wong & Oscar Hernandez
3.2.1.5	Early Fire Detection	The system reduces the propagation of a fire by minimizing false alarms and notifying the user or fire department as soon as a fire is detected.	Untested	All



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Thank you for your attention!

Questions?